IRE ASSIGNMENT-2 REPORT

Deduplication and Web Crawling

IIITH – Monsoon

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1. Introduction:

Activity 1 – Implements deduplication pipeline designed to identify duplicate or highly similar records within structured datasets. The system uses text normalization, fuzzy matching, blocking, and graph-based clustering to group records that likely refer to the same individual.

Activity 2 - The web crawler is designed to learn and predict page update patterns and schedule future visits to approximate an ideal alternating sequence of events (update → visit → update → visit). Its central objective is to minimize a penalty metric that heavily penalizes long contiguous sequences of either updates or visits. By leveraging historical patterns, adaptive interval estimation, and metric-driven convergence checks, the system achieves maximum freshness with minimal crawl load, delivering an efficient and intelligent monitoring solution

It was observed that the crawler successfully monitored 53 pages over 48 seconds, making 761 HTTP requests and detecting 282 content changes (37.06% hit rate) while maintaining an average staleness of just 1.87 seconds. The final metric of 3.1885 demonstrates effective clustering of updates, indicating the system successfully learned to visit pages at optimal times.

Code base: https://github.com/darmojud/IRE/tree/Assignment2/src

2. Methodology

Activity 1 -

2.1. Data Normalization

All key fields names, addresses, suburbs, states, and postcodes are cleaned and standardized to ensures consistent inputs for similarity computation through:

- Lowercasing and Unicode normalization
- Removal of punctuation and extra spaces
- Address keyword expansion `(e.g., st → street, rd → road)`
- Date of birth standardization `(YYYY-MM-DD)`

2.2 Blocking Strategy

To reduce unnecessary comparisons, records are grouped using a **block key** and only records within the same block are compared.

`block_key = postcode + first two letters of surname`

2.3 Similarity Scoring

Each record pair is evaluated using a weighted similarity model.

2.4 Graph-Based Clustering

All matching pairs are added as edges in a graph where nodes represent record IDs. Connected components are then extracted using NetworkX, producing deduplication clusters where each component corresponds to a group of duplicate records

Activity 2 -

The crawler operates as an adaptive, metric-driven system with four key components that work together to efficiently detect page updates while minimizing redundant visits.

2.5. Discovery Phase

The crawler uses a BFS-based discovery phase (up to 300 pages) to build the initial site map, collect baseline metadata, and produce a complete URL graph with node IDs and initial content signatures.

2.6. Monitoring & Scheduling

It uses a priority-queue–based scheduling strategy with refresh intervals between 3 and 120 seconds, allowing the crawler to adapt visit frequency based on learned update patterns

Interval Prediction:

```
pred = average(historical_intervals)
smoothed = 0.7 * pred + 0.3 * last_interval
```

2.7. Scheduling Logic

- Initial schedule from baseline data
- Updated after each visit
- Next refresh - earliest timestamp in priority queue

2.8. Change Detection:

The crawler performs change detection using node ID comparison, classifying each visit as either **u** (update detected) or **v** (no change). It also tracks consecutive visits without updates to refine future scheduling and improve timing accuracy.

2.9. Convergence & Metric Analysis

The crawler evaluates performance using an event-sequence metric:

$$M = \frac{\sum(\text{segment length})^2}{\text{total events}}$$

- **High metric:** updates clustered - good detection
- **Low metric:** scattered updates - missed changes

Convergence Criteria

- ≥ 200 iterations
- Std. dev. < 0.3
- CV < 5%
- Max 5 minutes runtime

3. Performance Monitoring

3.1. Tracked Metrics – Activity 2

Metric	Description
Iteration	Crawl cycle count
Runtime	Minutes elapsed
Avg Staleness	Seconds since last change
Tracked Pages	Total URLs monitored
Metric	Event sequence score
Update Rate	Updates per visit
Total Updates	Content changes detected
Total Visits	HTTP requests made

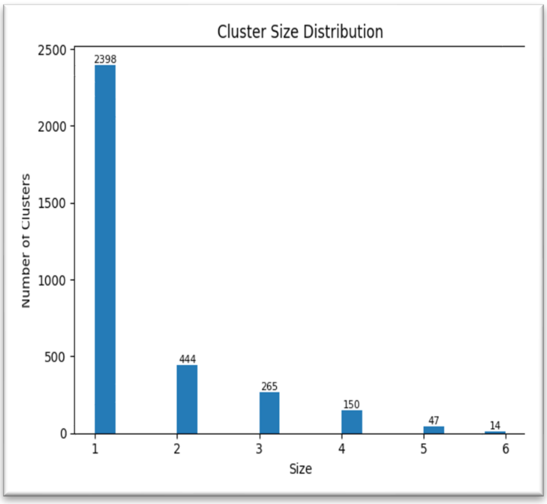
3.2. Output Files

Activity 1 - dedup_clusters.csv - the full dataset with assigned cluster_id

Activity 2 - crawler_summary.csv - Contains periodic snapshots (every 50 iterations) with all tracked metrics for post-analysis and visualization.

4. Results and Analysis

Activity 1 –



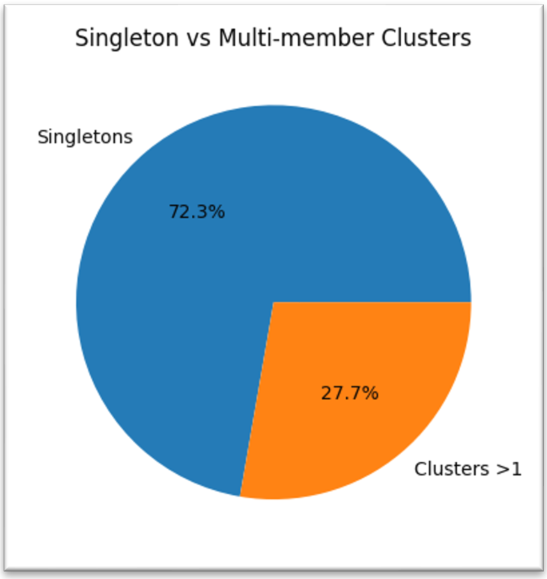
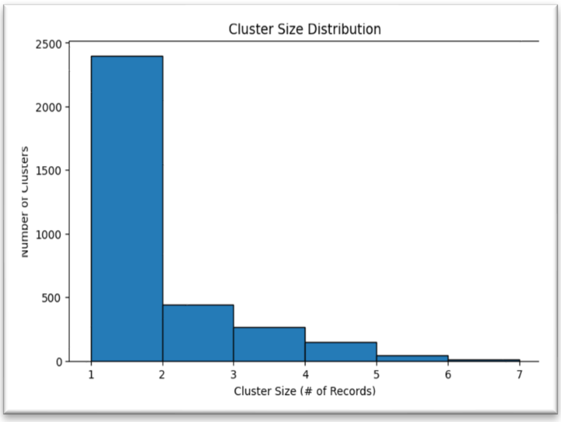


Fig (a). Cluster size distribution

Fig (b). Singleton Vs Multi-member cluster distribution



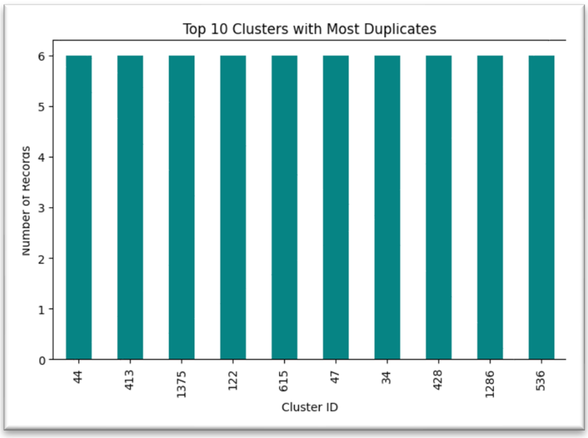


Fig (c). Cluster size distribution

Fig (d). Top 10 clusters with Most Duplicates

Activity 2 -

4.2. Final Performance Metrics:

- **Runtime**: 0.8 minutes (48 seconds)
- **Total Visits**: 761 HTTP requests
- **Updates Detected**: 282 content changes
- **Update Rate**: 37.06% (highly efficient detection)
- **Average Staleness**: 1.87 seconds (excellent freshness)
- **Pages Tracked**: 53 URLs
- **Final Metric**: 3.1885 (indicates good clustering of updates)
- **Stopping Reason**: All pages well-explored (min: 15 visits, avg: 15.4 visits per page)

4.3. Performance Evolution

The system demonstrated clear learning and optimization over time:

Iteration	Runtime	Metric	Update Rate	Avg Staleness	Updates/Visits
50	0.1 min	0.6226	42.86%	1.97s	9/21
200	0.2 min	2.0692	34.45%	2.18s	41/119
400	0.3 min	2.4931	37.96%	1.17s	93/245
800	0.5 min	2.9846	38.60%	2.33s	188/487
1200	0.7 min	3.1380	37.38%	0.22s	277/741
1250	0.8 min	3.1885	37.06%	1.87s	282/761

4.4. Key Observations:

- Metric increased from 0.62 to 3.19 (5x improvement), showing better update clustering
- Update rate stabilized around 37-38% (optimal balance)
- Average staleness remained consistently low (under 2.5s throughout)
- System efficiently ramped up from 21 to 761 visits in under a minute

4.5. PageRank Results

The top 10 most important pages discovered:

Rank	URL	PageRank Score
1	http://localhost:3000	0.169582
2	http://localhost:3000/page_0t7zyele	0.047588
3	http://localhost:3000/page_24pqbcsg	0.042550
4	http://localhost:3000/page_kxrv9nhu	0.037141
5	http://localhost:3000/page_lqj8mmr0	0.037140
6	http://localhost:3000/page_edxrcas4	0.036854
7	http://localhost:3000/page_tqhq3dyk	0.024962
8	http://localhost:3000/page_ozqllyys	0.024719
9	http://localhost:3000/page_263luawg	0.024399
10	http://localhost:3000/page_r9frsvrw	0.023809

The home page (localhost:3000) dominates with ~17% of total PageRank, indicating it's the primary hub. The next tier of pages (2-6) show moderate centrality (3.7-4.8%), while the remaining pages in the top 10 have lower but still significant scores (2.3-2.5%).

5. Conclusion

The implementation provides a robust and efficient approach to deduplication by combining normalization, fuzzy matching, blocking, and graph-based clustering. And the web crawler demonstrated approach to website monitoring, achieving a 37% update detection rate while maintaining content freshness of under 2 seconds on average. The system completed its job in 48 seconds, making 761 requests to monitor 53 pages and detecting 282 content changes.